



Designing High-Performance Fuels through Graph Neural Networks for Predicting Cetane Number of Multicomponent Surrogate Mixtures

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Abstract

Cetane number (CN) is an important fuel property in designing high-performance fuels in recently diversifying compression ignition engines. We introduce graph neural networks (GNNs) that predict CNs of multicomponent surrogate mixtures when only 2D structures and mole fractions of molecules are given. It considers the influences of

mixing multiple components and their chemical structures on CN, reproducing the non-linear blending behavior observed for certain mixtures. We trained the GNNs using the CNs of 1,143 mixtures, and reliable accuracy was achieved with mean absolute errors of 3.4–3.8 from the cross-validation. Lastly, we analyzed the chemical structural effects on non-linear blending behavior.

Introduction

As the regulations on the emission from medium-to-heavy duty vehicles get stringent, the combustion strategies in internal combustion (IC) engines get diversified, requiring a high-performance fuel (HPF) tailored to each application. The HPF indicates the fuels with desirable autoignition and vaporization characteristics for maximizing the thermal efficiency, maximum load, and combustion stability of a specific type of internal combustion engine. Cetane number (CN) has been considered an important fuel property in designing HPFs for advanced/conventional compression ignition engines as it is determined by both the chemical reactivity and vaporization characteristics of fuel.

For example, partially premixed compression ignition (PPCI) engines need low-reactivity fuels (low CN) to increase the fuel-air mixing time, which effectively reduces both soot and nitric oxide emission[1]. On the other hand, reactivity-controlled compression ignition (RCCI) engines require a high CN fuel to ignite the ambient fuel-air mixture with low-reactivity, typically natural gas, ethanol, or ammonia[2].

In this regard, *a priori* prediction of CN has been of particular interest to both the automotive and refinery

industries. One old study from Gülder and Glavincevskl[3] correlated the CN with the number of different carbon types identified from the proton nuclear magnetic resonance (¹H NMR) spectroscopy. The derived correlation successfully predicted the CN of 67 diesel fuels with a deterministic coefficient (r^2) of 0.992. This finding stimulated further efforts in the CN prediction based on molecular structural features such as the number of double bonds and carbon atoms[4]. The group additivity model is one of the most comprehensive ways to correlate the molecular structure and the CN, while the contribution of each molecular fingerprint is still under investigation[5]. A recent paper from our research group also utilized the group additivity for estimating the CN of ether-containing fuels based on their key molecular features (number of primary, secondary, tertiary, and quaternary carbons adjacent to or far from oxygen atoms), showing a remarkable accuracy with mean square error (MAE) of 6.14 on 55 saturated linear, branched, and cyclic ethers[6].

Meanwhile, machine learning (ML) technologies are being recognized as a promising way of predicting CN. Santana et al.[7] provided one of the earliest examples of applying ML in CN prediction. Their model was based on the

multilayer perceptron with back-static propagation, reproducing the experimental CN of 147 hydrocarbon fuels with a standard error of 8 CN units. The ML model was further exploited to cover the oxygenated fuels several years later by Saldana et al.[8], where the final model combining the feed-forward artificial neural network, support vector machines, and partial least squares regression showed the MAE of 4.7 on the 229 CN data points of hydrocarbons and oxygenates. Follow-up studies from other research groups[9, 10, 11, 12] have increased the size of the training dataset while keeping the prediction accuracy within a reasonable bound (~5 CN units). As an example, our previous study[13] applied the graph neural networks (GNN) for CN prediction to leverage the successful results from the group additivity model. With a proper evaluation of uncertainty in the database, the GNN predicted the CN of single compound fuels with an MAE of 2.4 CN units on the biggest dataset ever (630 single compound fuels), showing the competence of GNN in CN prediction.

Despite the recent progress in the CN prediction for single compound fuels, those for multicomponent fuels are still a challenging problem. Most prediction models for multicomponent fuels require the fuels' physical properties as input variables[14], which is not viable in the *de novo* design of HPFs. Otherwise, the CN of multicomponent fuels is estimated as the linear combination of each component's CN, so-called the linear mixing rule[15, 16]. However, the linear mixing rule of CN can sometimes fail due to the complicated synergistic effect between individual components in the fuel. The recent study from Fioroni et al.[17] suggested that the synergistic effect for research octane number (RON; inversely correlated with CN[18]) originated from the OH radical scavenging of low-reactivity components. Their simulation results indicate that the fuel that forms a resonance stabilized radical through OH addition reaction is the most effective in inducing the synergistic effect in RON of multicomponent fuels, which suggests the strong dependency of the blending nonlinearity on the structural features of individual components.

As an effort to resolve this complicated nature of multicomponent fuels' CN prediction, a previous study by Basu et al.[19] utilized an artificial neural network for predicting the CN of 60 multicomponent diesel fuels based on their structural parameters observed from ^1H NMR. They divided the spectra from ^1H NMR into 17 regions to represent the structural features from different molecular groups (*n*-paraffins, *iso*-paraffins, olefins, cycloalkanes, and aromatics), based on which the ANN model shows reasonable accuracy for both training and validation set ($r^2 = 0.91$ and 0.85 , respectively). Recently, Dalmiya et al.[20] applied convolutional neural networks (CNNs) in predicting the multicomponent fuel's CN. They measured the infrared spectra of multicomponent fuels on $2400 - 800\text{ cm}^{-1}$ to characterize the molecular fingerprints of individual components. The measured spectra were directly used to train the CNN model and predict the CN of multicomponent fuel, which showed remarkable accuracy in reproducing the experimental CN of 88 single compounds and multicomponent fuel (mostly jet fuels) with R^2 over 0.93. These two studies suggest the machine learning model is a promising way to connect the molecular fingerprints of individual components to the CN of multicomponent fuels.

In this study, we suggest a novel structure using GNN to predict the CN of multicomponent fuels. To do so, we assemble the largest database of experimental CN measurements to date. Then, using a linear approximation of nonlinear blending behavior with a learned weight matrix, our novel model can accurately predict the CN of fuel blends containing up to 12 distinct chemical species. Here, we present the structure of the model and analyze its performance on a subset of the compiled dataset. We also present a detailed analysis of the influence of the weight matrix in approximating nonlinear blending behavior. It is our hope that the model will serve as a valuable tool for fuel scientists to formulate surrogate blends with specific CNs and thus, ignition properties.

Methods

Building The Database

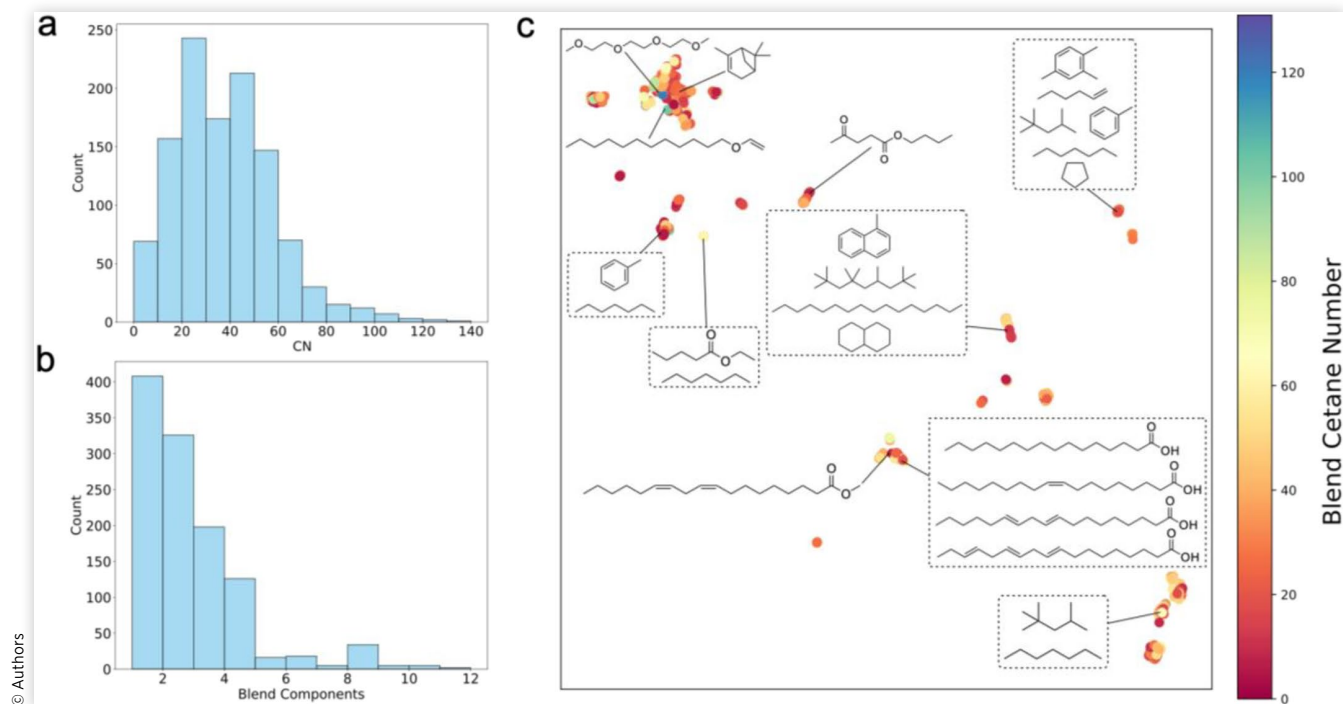
The chemical space from which fuel compounds are selected from is seemingly small: compared to databases used for, say, the design of novel pharmaceutical chemotypes, the molecular diversity of compounds used in fuel blends is much smaller[21]. However, even this subspace contains a great variety of chemical compounds, with unique structural motifs. The creation of models that do not adequately treat the breadth of this subspace has hindered commercial exploitation of the full range of possible fuel properties. In fact, without the ability to make confident predictions over the full fuel chemical space, models can restrict the palette of compounds available to fuel designers.

It is with this in mind that we have created the largest and most diverse database of measured CNs to date. Not only is a wide range of CNs represented in our database, but the associated blends consist of a diverse array of structural motifs, thus representing a wide range of combustion behaviors. Prior work regarding the prediction of CNs for pure fuel compounds used a tiered database, with an increase in tier number indicating a reduction in measurement data quality confidence[13]. The most-confident measurements consisted of 443 data points, with a total database size of 630 compounds. These 443 data points were used as a starting point to build the database; to these, high-quality measurements of the CNs of fuel blends were added, resulting in a total database size of 1143 measurements with CNs ranging from 0 to 131 (Figures 1a and 1b). A majority of the measured CNs fall under 70, with the mean and median values being 37.9 and 35.8, respectively.

To better characterize the diversity of the chemical compounds comprising the database, clustering using structural fingerprints was performed. The Morgan fingerprint is a "circular fingerprint" that seeks to accurately represent molecular structures via repeated substructure matches[22]. Each molecule is assigned a vector representation representing unique substructure environments; these generated vectors can then be compared using similarity scores such as the Tanimoto or Dice similarity.[23, 24]

Initially, the minimum length of vector representation was found to reduce computational overhead. It was found

FIGURE 1 (a) Distribution of cetane numbers across database components (b) Distribution of number of blend components (c) UMAP embedding of component Morgan fingerprints. For database components comprised of more than one fuel compound, a bitwise OR operation was performed on each compound vector, yielding a single 32-bit vector for each database component.



that a 32-bit Morgan fingerprint with radius 2 was sufficient to distinguish between a blend of propane and butane, and a blend of butane and pentane. Hence, a 32-bit Morgan fingerprint for each fuel component was calculated and a binary OR operation applied across fuel components in the same mixture, thus resulting in a 32-bit vector containing structural information about all blend compounds. This 32-dimensional vector was then subject to uniform manifold approximation and projection (UMAP) embedding for visualization of the chemical spaces occupied by each blend compound (Figure 1c)[25]. It is observed that several distinct clusters can be seen, with oxygenates, hydrocarbons, and respective blends forming disparate groupings in chemical space. This reflects the diversity of fuel compounds contained in the database.

Development of the Graph Neural Networks

We built our GNN by employing Python 3.7[26] with the TensorFlow 2.5 library[27]. The neural fingerprint (NFP) library[28] was also utilized for implementing the deep learning with the message-passing GNN. The input atom, bond, and global features of molecules were generated from their 2D structures, using the RDKit cheminformatics package[29]. Detailed description of GNN architecture is available in the next section (*vide infra*). The database, source code of the GNN, and trained weights are available through GitHub[30].

Results and Discussion

Nonlinear Blending Behavior

Fuel blends containing chemically diverse constituents often show nonlinear blending behavior, i.e., the CN of a fuel blend need not be a simple linear combination of the CN of its constituents. Since this behavior may be caused by a variety of factors, the interactions are difficult to model in a generalizable manner with a single, closed form expression. For example, as the concentration of ethanol is increased in a fuel blend consisting of ethanol and diethyl ether, the CN exhibits a non-linear decrease (Figure 2), in contrast to a linear fit of the experimental data. It should be noted that the linear fit referenced does not account for prior knowledge of pure compound cetane numbers, and instead seeks to fit the full range of blending behaviors across constituent mole fractions. A recent theoretical study has shown that the stability of the transition state involved in hydrogen abstraction and radical formation from fuel molecules is enhanced by hydrogen bonding interactions, thus enhancing the rate of reaction and increasing reactivity/CN[31].

On the contrary, a mixture of isooctane and tetralin exhibits first decreasing, then increasing cetane number as the isooctane concentration is increased (Figure 3). This divergent behavior can be explained by the effects of dispersive Van der Waals interactions between the two compounds; in addition, the cleavage of a carbon-carbon bond can result in the formation of a highly stable tertiary butyl radical per

FIGURE 2 Nonlinear antagonistic blending behavior of ethanol-diethyl ether fuel blends. The experimentally observed cetane numbers can be reproduced by using a nonlinear approximation for the blending behavior.

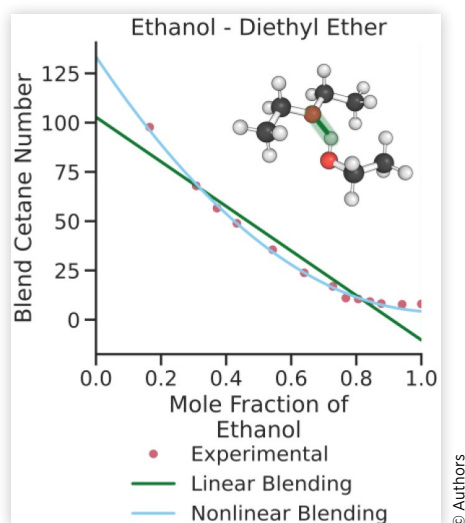
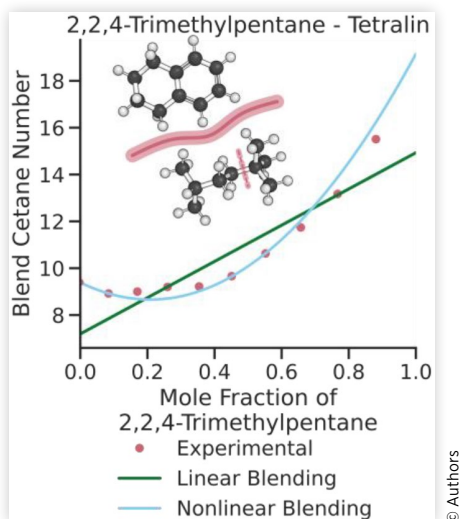


FIGURE 3 Nonlinear antagonistic blending behavior of 2,2,4-trimethyl pentane (*iso*-octane) fuel blends. Dispersive Van der Waals interactions and kinetic factors may play a part in the observed experimental nonlinearity.



inference using the ALFABET BDE prediction tool [32], accelerating the rate of downstream combustion reactions.

In addition, nonlinear blending behavior can be caused by the influence of downstream oxygenated intermediates. Recent work has shown that several competing pathways involving reactions of fuel species with the highly reactive peroxy radicals can lead to branching and chain propagation reactions, with the net effect of increasing reactivity at lower temperature regimes. These reactions involve the formation of a ketohydroperoxide intermediate and a hydroxyl radical[32]. On the contrary, at higher temperature regimes, oxygen addition is inhibited, and β -scission reactions become more prominent, leading to the observance of negative

temperature coefficient behavior[32]. This complex variation of observed properties thus induces nonlinear blending behavior.

Thus, the problem we are faced with is twofold: any predictive model for blend CNs must incorporate both intramolecular information about the chemical kinetics, and information about intermolecular interactions. Here, we treat the intramolecular chemical kinetics using the GNN embedding; previous work has shown that the molecular embedding utilized here is capable of learning the bond dissociation enthalpy of any bond in a molecule[33]. Since the bond dissociation enthalpy represents an energetic barrier to the slow unimolecular hydrogen abstraction step that constitutes the initial reaction of a fuel molecule during a combustion process, the neural network input embedding adequately captures the kinetics of the reaction. Details about our GNN architecture are explained in the following sections.

Formulation for Modeling Nonlinear Blending Behavior

As outlined in Figures 2 and 3, some mixtures exhibit nonlinear relationships between CNs and mole fractions of constituent molecules, and the predictive model should capture these trends. To this end, the governing equation regarding the description of mixtures was established and used in the ML model.

Such formulation should be consistent with underlying physics and thermodynamics regarding mixtures. Of note, the deviations from the linearity are analogous to the non-ideal mixing behavior of vapor pressures of liquid mixtures. As shown in Figure 4, the extended Raoult's law formulates mixture's vapor pressure ($P_{mixture}$) as a function of mole fraction (χ_i), activity coefficient at infinite dilution (γ_i), and vapor pressure ($P_{pure,i}$) of each component[34, 35, 36].

FIGURE 4 Description of the assumptions and formulations for modeling non-linear blending behavior.

- Vapor pressure of mixtures:**

$$P_{mixture} = \sum_i (\chi_i \gamma_i P_{pure,i})$$
- Likewise, cetane number of mixtures:**

$$CN_{mixture} = \sum_i (\chi_i \gamma_i CN_{pure,i})$$

where $\gamma_i CN_{pure,i} = CN_i = \sum_j (\chi_j w_{ij})$,
 $(\chi_i$: Mole fraction, γ_i : Activity coefficient at infinite dilution)

cf. Linear regression ($\gamma_i = 1$)

$$CN_{mixture} = \sum_i (\chi_i CN_{pure,i})$$
- Example: Ethanol – diethyl ether mixture**

$$W = \begin{bmatrix} 5.8 & 8.4 \\ 6.7 & 136.9 \end{bmatrix} \begin{array}{l} - CN_{pure,i}: \text{ethanol, diethyl ether} \\ - \text{Infinite dilution CN of:} \\ \quad \text{Ethanol in diethyl ether} \\ \quad \text{Diethyl ether in ethanol} \end{array}$$

* $i = 1$: Ethanol, $i = 2$: Diethyl ether,
 $\chi_1 = 0.64$, $\chi_2 = 0.36$, $CN_{mixture} = ?$
 $CN_1 = 5.8 \cdot 0.64 + 8.4 \cdot 0.36 = 6.71$
 $CN_2 = 6.7 \cdot 0.64 + 136.9 \cdot 0.36 = 53.59$

 $CN_{mixture} = 6.71 \cdot 0.64 + 53.59 \cdot 0.36 = 23.59$
 (cf. Exp.: 23.8, Linear regression: 30.4)

The activity coefficient γ_i quantifies the deviation from the linearity; if it is 1, the equation corresponds to a simple linear model for ideal liquid solutions. Otherwise, it describes non-linearity.

Here, we hypothesize that the similar equation can also be applied to the CN of mixtures (CN_{mixture}), and accurate prediction of γ_i leads to better description of the non-linear mixing behavior. CN_{mixture} can be predicted by calculating the summation of $\chi_i \gamma_i CN_{\text{pure},i}$ values of each component (Figure 4). Inspired by the Wilson equation[37], $\gamma_i CN_{\text{pure},i}$ term can be equated to the summed value of $\chi_j w_{ij}$, corresponding to a CN contribution of component i in a given mixture (Abbreviated as CN_i). Each row of the **W** matrix accounts for intermolecular interactions when one component i is infinitely diluted with respect to others (j).

One example of **W** matrix is shown in Figure 4, for the ethanol – diethyl ether mixture. The diagonal elements ($w_{11}=5.8$, $w_{22}=136.9$) indicate CNs of pure ethanol and diethyl ether determined by the regression. The infinite dilution of ethanol in diethyl ether, and vice versa, is considered by the off-diagonal elements ($w_{12}=8.4$, $w_{21}=6.7$). It is noteworthy that w_{12} and w_{21} are independent to the concentration of ethanol and diethyl ether. Then, CN contributions of each compound (CN_1 and CN_2) are evaluated by the linear combination of each element in a row multiplied by a mole fraction. Lastly, CN_{mixture} is calculated by a mole-fraction- weighted sum of CN_1 and CN_2 . We emphasize that the influence of each component's mole fraction is considered twice, based on the analogy with activity coefficients. This is because the activity coefficient itself is affected by mixture's composition, so as CN_{mixture} is.

It should be noted that **W** can be asymmetric due to different infinite dilution effects from each component, which is useful for modeling the non-linear blending behavior. The formulation with the **W** matrix well reproduces the experimental CN of the mixture with 64% ethanol and 36% diethyl ether (23.59 and 23.8 from the regression and experiment, respectively). This estimation is more accurate than the linear regression which resulted in the CN of 30.4, ultimately leading to a better description of non-linear blending behavior than linear regression (Figure 4). Previous studies also adopted the similar approaches; they developed the reliable composition-based predictive models for octane and cetane numbers[16, 38] based on the analogy of CNs of blends with activity coefficients. A root-mean-square error of 1.25 was achieved for CNs of 203 diesel fuel mixtures[16]. Here, we pursued the development of predictive models with broad applicability by using more extensive database consisting of 1,143 data points and modeling non-linearity as a function of molecular structures as well as mixture's composition.

Baseline Analysis for the Predictive Model Development

Before developing the actual ML model, baseline model analysis was performed for the model's feasibility. Specifically, we validated our hypothesis regarding the formulation shown in Figure 4 for modeling non-linearity. The whole database

consisting of 1,143 was used for the linear regression and the regression with **W** matrix, based on the equations shown in Figure 4. Adopting the **W** matrix showed a lower mean absolute error (MAE) of 0.74 compared to the linear regression (MAE: 1.86, Figure 5). The higher accuracy obtained from the **W** matrix demonstrates the feasibility of predicting non-linear blending behavior by considering infinite dilution effects, in analogy to the equations for activity coefficients.

However, the size of **W** matrix grows proportionally to the number of components (N) in a mixture; it can be up to 12x12 within our database, and potentially even larger for mixtures with more than 12 components. For efficient regression, we adopted a low-rank approximation (Figure 6) of **W**. The ($N \times N$) matrix **W** is obtained through a dot product of two matrices **U**^T and **V** whose sizes are ($N \times r$) and ($r \times N$) respectively. The rank, r , can be adjusted so that the number of regressed parameters is minimized while maintaining the accuracy of **W**. The direct evaluation of **W** requires the adjustment of N^2 parameters, whereas low-rank approximations can reduce this number to $2rN$. More specifically, for 12-component mixtures, there are $12^2=144$ parameters in **W**, and the number of parameters is reduced if r is set to lower than six ($2rN=2 \cdot 6 \cdot 12=144$).

To find an optimal number of r , we repeated the regression with **U** and **V** matrices with varying r . Figure 6 illustrates the MAEs from the regressions at different r . $r=1$ showed a MAE of 2.54, which is worse than that from linear regression, but a MAE of 0.82 was achieved at $r=2$. In other words, the rank of 2 is sufficient to obtain the accuracy comparable to that from the full **W** matrix (MAE: 0.74). Increasing r to 3 and 4 slightly improved the accuracy with the MAEs of 0.78 and 0.75, respectively, but at the cost of computational expense. The computational cost of regression is proportional to the number of regressed parameters ($2rN$), so the regression at $r=3$ and 4 are 1.5 and two times more expensive than that at $r=2$, respectively. Therefore, it can be concluded that $r=2$ is an optimal rank value that can achieve the best compromise between accuracy and computational costs.

The insights gained from the above baseline analysis were utilized in building graph neural networks (GNNs) for predicting CNs of mixtures with incorporating the **W** matrix. The GNNs are for embedding molecular structures of each

FIGURE 5 Results from the analysis of two baseline models: (1) Linear regression of mixtures' CNs in terms of mole fractions and CNs of each component. (2) Regression with using the **W** matrix.

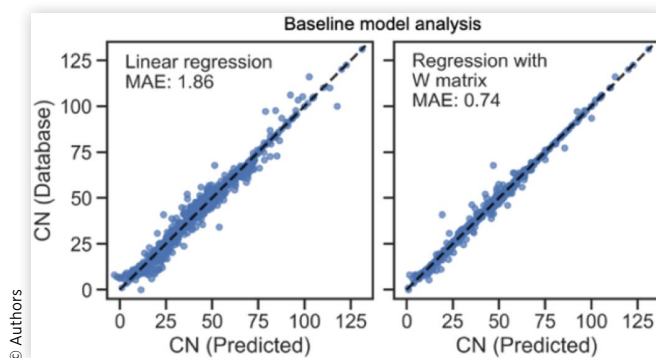
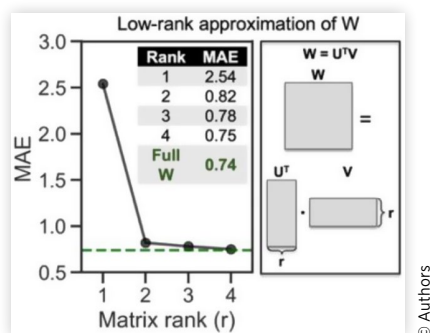


FIGURE 6 The low-rank approximation of $\mathbf{W}$ and accuracies from the regression with varying the rank r .



component and considering ‘intramolecular’ structural effects on CN. The subsequent part of the model accounts for the ‘intermolecular’ influences on CN by constructing $\mathbf{W}$ with the $r=2$ dimensionality reduction. Details are discussed in the next section.

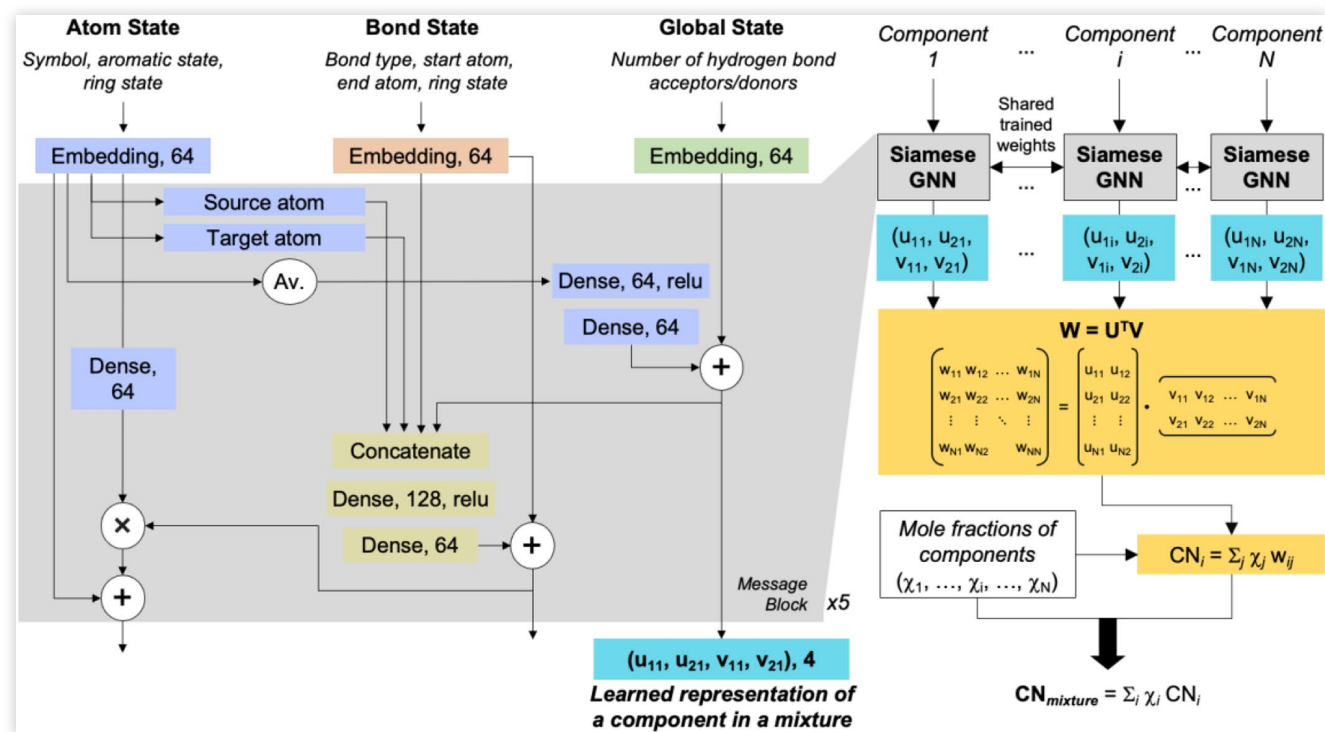
Graph Neural Networks And Molecular Representations

Figure 7 illustrates the architecture of our GNN. The first part consists of N parallel message-passing neural network (MPNN) blocks for encoding individual component molecules in a mixture and accounting for their chemical

structural effects on CN. Of note, the model was trained to predict CNs of mixtures consisting of up to 12 components, as 12 is the maximum number of components in our curated database. However, in principle, the model architecture can be readily modified and expanded to any numbers of components. Two-dimensional molecular structures are given as inputs, with their atom and bond features. The model also inputs two global features (The number of hydrogen bond donors and acceptors: N_{HBD} , N_{HBA}). Such features influence fuels’ physicochemical properties pertinent to CN, such as heat of vaporization, boiling point, reactivity, enthalpy of mixing, and non-linear mixing behavior of CNs (Figures 2 and 3)[13, 39, 40].

All these features are embedded as vectors and undergo message layers in which mathematical operations among the vectors are carried out. Passing through multiple layers enables the model to learn local and non-local environments of atoms and bonds, and thus, their impacts on CN. The number of message-passing layers was set to five (Figure 7), as the MPNN with five layers showed the best accuracy for CNs of 630 pure compounds[13]. All the MPNN blocks share the trained weights, akin to Siamese neural networks, so that the whole GNN is trained consistently with respect to various compounds in the database. We successfully applied the similar MPNN architecture to predicting CNs of pure fuel compounds including 630 hydrocarbons and oxygenates, with an overall MAE of 5.34 from the 10-fold cross-validation[13]. While the model for pure compounds has only one MPNN

FIGURE 7 Architecture of our graph neural networks (GNNs) for predicting cetane numbers of mixtures. Each component molecule passes through the GNN consisting of five message-passing layers. Each layer outputs the updated atom, bond, and global state vectors, which are the inputs for the next layer, as indicated by the three arrows penetrating the message block. The final global state block outputs a four-dimensional vector of which elements are used to build the $\mathbf{W}$ matrix and account for the non-linear mixing behavior.



block and outputs the CN, here, each of MPNN blocks outputs a four-dimensional vector for component i (u_{1i} , u_{2i} , v_{1i} , v_{2i} , Figure 7) that is used for modeling multicomponent interactions. These vectors from the GNN blocks correspond to learned representations of a component in a mixture.

The dimension of four was chosen because the rank of 2 was found to be optimal in the baseline model analysis (Figure 6) for the low-rank approximation of $\mathbf{W}$ by multiplying two matrices ($\mathbf{U}$ and $\mathbf{V}$). The former and latter two elements comprise $\mathbf{U}$ and $\mathbf{V}$, respectively: (u_{1i} , u_{2i}) and (v_{1i} , v_{2i}) for component i . These four-dimensional vectors are collected from all MPNN blocks and then concatenated, resulting in two ($N \times 2$) matrices for rank-2 approximation: $\mathbf{U}=[(u_{11}, u_{21}), (u_{12}, u_{22}), \dots, (u_{1N}, u_{2N})]$ and $\mathbf{V}=[(v_{11}, v_{21}), (v_{12}, v_{22}), \dots, (v_{1N}, v_{2N})]$, where N is the number of components. Next, $\mathbf{U}$ and $\mathbf{V}$ are multiplied to obtain $\mathbf{W}=\mathbf{U}^T\mathbf{V}$. The elements of $\mathbf{W}$ are trained and adjusted so that it can account for the infinite dilution effects when one component is mixed with the other, and vice versa.

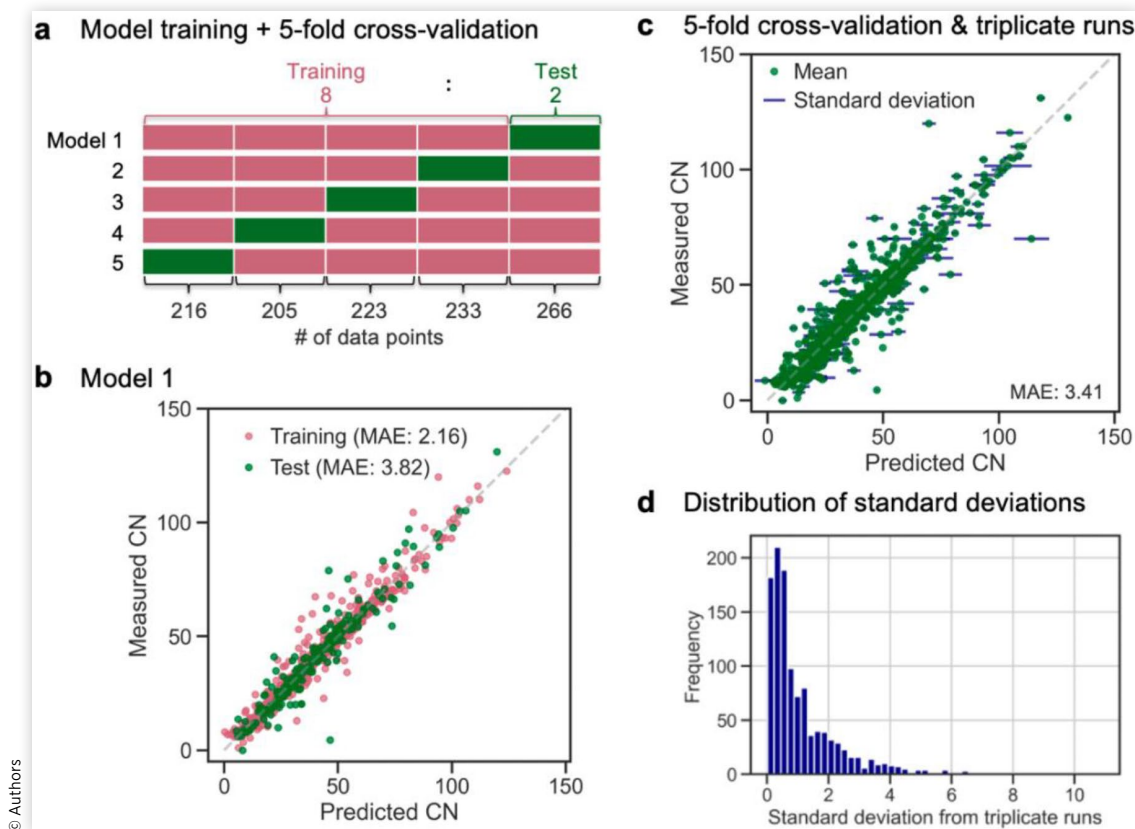
In summary, a multi-partite graph neural network was constructed, with parallel GNNs producing independent representations of each blend component molecule. The $\mathbf{W}$ matrix associated with rank-reduced approximations is then learned in conjunction with the molecular representations to predicted CNs of mixtures.

Model Training and Evaluation of the Model Accuracy

The GNN depicted in Figure 7 was then trained, and its accuracy was evaluated. The database was split into five parts for the 5-fold cross-validation (Figure 8a). Each fold consists of 216, 205, 223, 233, and 266 data points, respectively. The numbers of data points are not identical among the folds, since each mixture has different number of data points with varying mole fractions. For example, in Figure 2, there are 13 data points for the ethanol – diethyl ether mixture, whereas for the iso-octane – tetralin mixture (Figure 3), 10 different compositions exist in the database. The five-fold data split was carried out multiple times with changing random seeds, and the random split was chosen so that the deviation of data points among the folds is minimized. Then, five models were trained with changing training and test sets (Model 1 – 5, Figure 8a). The training:test set ratio was kept approximately 80:20.

Figure 8b shows the parity plot for Model 1. The MAEs for the training and test set are 2.16 and 3.82, respectively. It should be emphasized that this accuracy is comparable to or higher than previous studies[3, 5, 8, 9, 10, 12, 13, 16, 19, 20], while the largest database was used (1,143 data points). Moreover, these MAEs are similar to the uncertainty of experimental CN measurements using standard devices such as CFR, IQT, and AFIDA. The typical reproducibility ranges of

FIGURE 8 (a) Data splits for the 5-fold cross-validation, with the number of data points for each chunk of data. (b) A parity plot depicting the accuracy of the training and test sets of Model 1. (c) A parity plot of test sets from three repetitions of the 5-fold cross-validation, with the standard deviations of three predicted CN values per each data point. (d) A histogram showing the distribution of standard deviations of three predictions for the whole dataset (1,143 data points).



these devices are 2.8-4.8, 1.6-2.6, and 1.5-2.3 CN units, respectively, for CNs of diesel fuels ranging from 40 to 56 [13, 41, 42, 43]. Model 1 was used for the subsequent analysis regarding how the model describes non-linear mixing behavior (*vide infra*). Furthermore, the 5-fold cross-validation was repeated for three times, and the accuracy of test sets from all folds were assessed, to confirm that the model is not susceptible to overfitting.

Figure 8c illustrates the results from the 5-fold cross-validation with triplicate runs, resulting in a total of 15 separate validation runs. Predicted CNs were obtained by averaging three CNs, and the standard deviation from three predictions were also plotted. Of note, in Figure 8c, there are no data points from the training set, and the whole database is considered as the test set through the 5-fold cross-validation. The MAE is 3.41, showing no significant increase in the error when only test sets are used for the model evaluation. We also analyzed the standard deviations of predicted CN values from triplicate runs (Figure 8d); Most of the standard deviations (92.5 %) do not exceed 3.0, which is also comparable to the experimental uncertainties discussed above.

Intpretation of the W Matrix Trained for Modeling Nonlinear Blending Behavior

Next, we use a per-blend additive approximation to linear mixing behavior via the use of a learned weight matrix to model the intermolecular interactions. Approximation of nonlinear behavior with linear transformations has previously been well established in the fields of dynamical systems and process modeling. For instance, von Neumann and Koopmans' formulation of the Koopman operator for the time evolution of classical systems has been rigorously studied. While it is possible to construct an infinite dimensional Hilbert space operator that accurately describes the dynamics of the system, computer systems are incapable of treating such infinite-dimensional operations. Thus, the task becomes one of finding a high-fidelity, finite dimensional operator that approximates the linear, infinite dimensional operator, which in turn approximates a nonlinear operator.

Here, we are faced with a similar goal: the approximation of an accurate finite-dimensional, low rank linear transformation that describes the nonlinear blending behavior observed when different fuel compounds are mixed. Unlike the Koopman operator problem, however, we are not faced with the issue of resolving the behavior of our system along an infinite number of dynamical modes; instead, we only need to treat mixing between blend components, thus constraining our linear operator (written in the form of the square weight matrix **W**) to the number of blend components, 12. To further reduce the complexity of the interaction behavior, and to aid in interpretation, the weight matrix was further subjected to a low-rank approximation; it was found that using a four-rank matrix for CN prediction yielded a mean absolute error of only 0.01 CN more than the full-rank matrix (Figure 6).

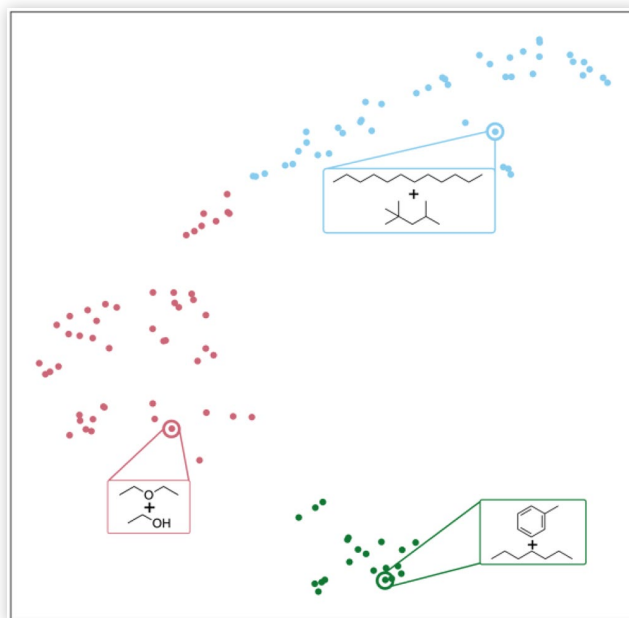
Since the weight matrices represent a concentration-independent means of characterizing intermolecular interactions, they were then subjected to further analysis, to further

investigate the non-linear nature of the blending interaction. Clustering and dimensionality reduction was performed using k-means clustering followed by Uniform Manifold Approximation and Projection (UMAP) embedding on the full rank learned weight matrices. It is observed that three distinct clusters are found (Figure 9); broadly, these correspond to blends that exhibit synergistic (cetane number higher than predicted by a linear model), antagonistic (cetane number lower than predicted by a linear model), and linear additive behavior. Since similar matrices are clustered together in the 12-dimensional latent space, the model can identify similar types of blending interactions and construct weight matrices accordingly.

This behavior was further analyzed by quantifying the degree of asymmetry in the learned weight matrices. As previously stated, the asymmetry in the weight matrices arises from infinite-dilution effects for each component – thus, the more asymmetric the weight matrix, the higher the deviation from linear blending behavior. Here, we study the degree of asymmetry for binary mixtures in the test set blends (binary mixtures were chosen to aid in interpretation of results). The degree of asymmetry is quantified by decomposing the matrix into purely symmetric and antisymmetric parts, and comparing the magnitude of each utilizing a Frobenius norm (written as $\|\bullet\|_{Frob}$) based symmetry score:

$$W_{sym} = \frac{1}{2}(W + W^T) \quad (1)$$

FIGURE 9 K-means clustering followed by 2D UMAP embedding of weight matrices for test set fuel blends. Clustering utilized 3 disparate clusters in the 12D latent space, corresponding to linear, synergistic, and antagonistic blending behavior respectively. The number of clusters used by the k-means algorithm was validated by silhouette score analysis (Appendix Figure A1).



$$W_{\text{antisym}} = \frac{1}{2}(W - W^T) \quad (2)$$

$$\text{Symmetry Score} = \frac{\|W_{\text{symm}}\|_{\text{Frob}} - \|W_{\text{antisym}}\|_{\text{Frob}}}{\|W_{\text{symm}}\|_{\text{Frob}} + \|W_{\text{antisym}}\|_{\text{Frob}}} \quad (3)$$

Blends with higher symmetry scores show linear mixing behavior, while those with lower symmetry scores show nonlinearity (Figure 10). For ease of interpretation, the symmetry scores were divided into three equally spaced broad classes: nonlinear blending (scaled symmetry score of 0-33), moderately linear blending (scaled symmetry score of 34-67), and linear blending behavior (scaled symmetry score of 67-100). As expected, hydrocarbon mixtures show greater symmetry, indicating tendencies toward linear blending behavior, while those mixtures containing oxygenated compounds show greater asymmetry and thus more nonlinear mixing behavior via intermolecular interactions. An example of hydrocarbon mixtures is shown in Figure 11: dodecane-propylbenzene mixtures with varying the mole fraction of dodecane from 0.0 (pure propylbenzene) to 1.0 (pure dodecane). For the mixture whose experimental CN is 44.5, the calculated CN using the weight matrix based linear approximation is 46.6, and the model predicted cetane number is also 46.6, indicating its linear blending behavior. The above analysis demonstrates that the governing equations for describing mixtures (Figure 4) are valid and the model can distinguish the fuel blends that show linear or non-linear blending behavior.

FIGURE 10 Scaled symmetry scores for binary mixtures in the test set fuel blends. Calculated symmetry scores were scaled to 100 for ease of interpretation.

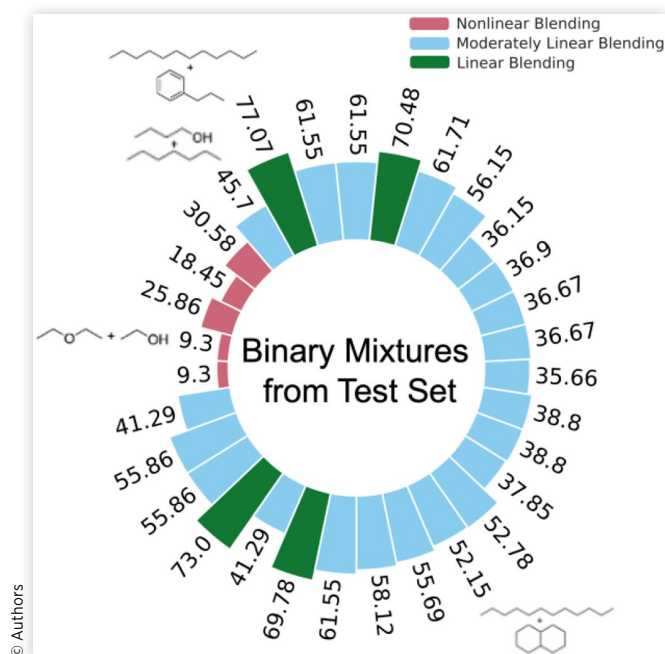
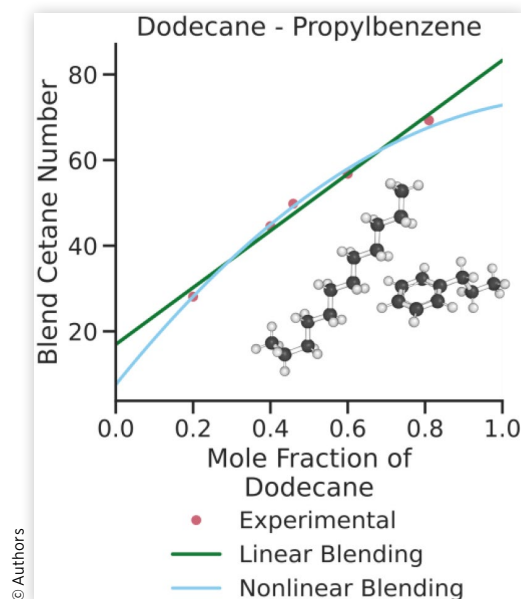


FIGURE 11 Linear blending behavior of the dodecane-propylbenzene fuel blend. By quantifying the asymmetry present in the learned weight matrix for this blend, this linear behavior can be predicted *a priori*.



Conclusion

We developed the GNNs that predict CN of multicomponent mixtures that can be fuel surrogates. Rapid predictions are possible (<1 second per one mixture when a CPU or GPU is used) as the model uses only the atom, bond, and molecular features generated from 2D structures. We achieved the high accuracy comparable to experimental uncertainties, with MAEs ranging from 3.4 to 3.8 from the 5-fold cross-validation. The standard deviation of predicted values did not exceed 3.0 for most cases when the model was trained three times. The model considers the effects of intermolecular interactions on CNs by adopting the formulation in analogy to the equations for activity coefficients in mixtures, leading to the prediction consistent with underlying physics and thermodynamics of mixtures. To further verify feasibility of the model, we found the relationship between the fuel's non-linear blending behavior and the values of trained weight matrix (W) in the hidden layer.

The GNNs for CN of mixtures manifest a successful expansion of our previous GNN for single compounds, by modeling both the influence of chemical structures and the infinite dilution of each molecule on CN. The model reliably reproduced the non-linear synergistic/antagonistic blending behavior of mixtures with respect to varying mole fractions of constituent molecules. Our approach will aid the design of new fit-for-purpose fuels whose CNs are either within or outside the range of typical diesel fuels (40-56). It can also be combined with predictive models for other fuel properties such as yield sooting index. Such integration of predictive models would enable the discovery of low-emission and high-performance fuels, responding to the recently increasing

demands on mitigating environmental impacts of vehicles while maintaining high engine performance.

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Appendix

FIGURE A1 Silhouette score plot for varied numbers of k-means hyperparameters. A larger score indicates better assignment of points to each cluster

